

Red Hat Lightspeed 1-latest

Assessing and monitoring security policy compliance of RHEL systems

Understanding the security compliance status of your Red Hat Enterprise Linux Infrastructure

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Abstract

Assess and track the security-policy compliance status of your Red Hat Enterprise Linux environment to determine compliance level and plan a course of action to resolve compliance issues.

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CHAPTER 1. RED HAT LIGHTSPEED COMPLIANCE SERVICE OVERVIEW

The Red Hat Lightspeed compliance service enables IT security and compliance administrators to assess, monitor, and report on the security-policy compliance of RHEL systems. Learn how to understand reporting, manage issues, and get the maximum value from the service.

The compliance service provides a simple but powerful user interface, enabling the creation, configuration, and management of SCAP security policies. With its filtering and context-adding features, IT security administrators can easily identify and manage security compliance issues in the RHEL infrastructure.

You can also create Ansible Playbooks to resolve security compliance issues and share reports with stakeholders to communicate compliance status.

1.1. REQUIREMENTS AND PREREQUISITES FOR THE COMPLIANCE SERVICE

Your Red Hat Enterprise Linux (RHEL) subscription includes the Red Hat Lightspeed compliance service, which you can use with all versions of RHEL that are currently supported by Red Hat.

You do not need additional Red Hat subscriptions to use Red Hat Lightspeed and the compliance service.

1.2. SUPPORTED CONFIGURATIONS IN THE COMPLIANCE SERVICE

Red Hat supports specific versions of the SCAP Security Guide (SSG) for each minor version of Red Hat Enterprise Linux (RHEL). To receive accurate compliance reporting, the system must have the supported SSG version installed.

The rules and policies in an SSG version are accurate for only one RHEL minor version and Red Hat Enterprise Linux minor versions ship and upgrade with the supported SSG version included. However, you might decide to use an earlier version temporarily. If a policy includes systems that use unsupported SSG versions, an **unsupported** warning with the number of affected systems is displayed next to the policy in Security > Compliance > Reports .

1.2.1. Frequently asked questions about the compliance service

How do I interpret the SSG package name?

Package names are in the format **scap-security-guide-0.1.43-13.el7** where, in this example, the SSG version is **0.1.43**, the release is **13**, and the architecture is **el7**.

What if Red Hat supports more than one SSG for my RHEL minor version?

When more than one SSG version is supported for a RHEL minor version, the compliance service will use the latest available version. This is the case for RHEL 7.9 and RHEL 8.1.

Why is my old policy no longer supported by SSG?

As RHEL minor versions get older, fewer SCAP profiles are supported.

What are other limitations of unsupported configurations?

Although you can still see results for a system with an unsupported version of SSG installed, the results might be inaccurate. The following factors apply with unsupported versions. Results for systems that use an unsupported version of SSG *are not included* in the overall compliance

assessment for the policy. Remediation plans are also not available for rules on systems that have an unsupported version of SSG installed.

Additional resources

- Red Hat Lightspeed Compliance - Supported configurations (Red Hat Knowledgebase)

1.3. BEST PRACTICES FOR THE COMPLIANCE SERVICE

To follow best practices and receive the most accurate results in the compliance service, make the RHEL operating system minor version visible and create security policies in the compliance service.

1.3.1. Make the RHEL operating system minor version visible

If the compliance service cannot access your RHEL operating system minor version, it cannot validate the supported SCAP Security Guide version and your reporting might not be accurate. With the insights-client, users can redact data such as the Red Hat Enterprise Linux operating system minor version from the data payload that is uploaded to Red Hat Lightspeed, but this redaction prohibits accurate compliance service reporting and is not recommended.

1.3.2. Create security policies in the compliance service

You can create your organization's security policies in the compliance service to do the following actions:

- Associate multiple systems with the policy
- Use the supported SCAP Security Guide for your RHEL minor version
- Edit which rules are included based on your organization's requirements

1.4. MANAGE USER PERMISSIONS FOR RED HAT LIGHTSPEED SERVICES

Manage user permissions to control access to Red Hat Lightspeed applications. Use the User Access feature to apply role-based access control (RBAC). Red Hat provides predefined groups and roles to help Organization Administrators manage user permissions.

1.4.1. User Access overview

You can simplify access management in the Red Hat Hybrid Cloud Console by using role-based access control (RBAC). The User Access feature assigns specific permissions to roles instead of individual users, which are then inherited when roles are assigned to user groups.

You can also create custom groups and roles to provide more fine-tuned control over specific features of Red Hat Lightspeed to suit the needs of your organization.

If you are an Organization Administrator, you can use the User Access feature under **Identity & Access Management → User Access** in the Hybrid Cloud Console to:

- Control user permissions and organize roles.
- Create groups that include roles and their corresponding permissions.

- Assign users to these groups, allowing them to inherit the permissions associated with their group's roles.

Additional resources

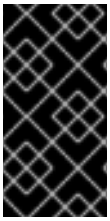
- User Access in the Hybrid Cloud Console
- User Access Configuration Guide for Role-based Access Control (RBAC)

1.4.2. Predefined groups in User Access

The two predefined groups available in User Access, **Default access** and **Default admin access** provide different baseline privilege levels for users in your organization. You can also create custom groups to align permissions with specific personas, job functions, or teams in your organization.

The Default access group

By default, the **Default access** group is assigned many granular predefined roles, so that group members have basic visibility. Because all users in your organization are members of the **Default access** group, they inherit all permissions assigned to that group. The **Default access** group is automatically updated by Red Hat.



IMPORTANT

If your Organization Administrator modifies the **Default access** group, for example, by removing roles to restrict access to specific applications or to use the consolidated roles, the group is automatically renamed to **Custom default access** Once converted, this group is no longer automatically updated by Red Hat.

The Default admin access group

The **Default admin access** group contains only users who have Organization Administrator permissions. This group is automatically maintained, and users and roles in this group cannot be changed.

The **Default admin access** group includes many (but not all) predefined roles that provide update and delete permissions. The roles in this group usually include **administrator** in their names.

TIP

To view the roles included in the **Default access** or **Default admin access** group, log in to the Hybrid Cloud Console, navigate to **Groups** in User Access, and select the group.

Additional resources

- User Access groups

1.4.3. Predefined roles assigned to groups

Understand how predefined roles in Red Hat Hybrid Cloud Console bundle permissions across multiple Red Hat Lightspeed applications to align with common user personas. Use predefined roles to reduce administrative effort, or create custom roles for more fine-tuned control over specific features.

The predefined roles are a starting point to help you control and manage user permissions. You can then use these roles to create custom roles that are tailored to your specific use cases and organization. For example, you can use the predefined granular roles to create custom roles that provide more fine-tuned

control over specific features of Red Hat Lightspeed.

By default, Red Hat provides a set of *consolidated roles* and a set of *granular* roles in the Red Hat Hybrid Cloud Console User Access UI. The consolidated roles significantly reduce the administrative effort required to manage user permissions, while the granular roles provide more fine-tuned control over specific features of Red Hat Lightspeed.

You can use the predefined consolidated and granular roles in User Access simultaneously, but using consolidated roles can significantly reduce the administrative effort.

Select from the predefined consolidated roles library

The Red Hat Hybrid Cloud Console provides three predefined, consolidated User Access roles to help you manage user permissions to Red Hat Lightspeed applications and services that run on registered Red Hat Enterprise Linux systems. These roles help simplify how the Organization Administrator creates groups and permissions for various levels of access to the Red Hat Lightspeed services. If you want to reduce the administrative effort required to manage user permissions and your use case aligns with the permissions included in these roles, select from the consolidated roles library.
The consolidated roles are as follows:

RHEL viewer: The *RHEL viewer* role provides users visibility without the ability to make changes. It allows read-only access to Red Hat Lightspeed. You can view system configurations, compliance reports, inventory data, patch information, vulnerabilities, and overall resource states and activities. The only action permitted with this role is to generate activation keys.

RHEL operator: The *RHEL operator* role allows active management of your Red Hat Lightspeed environment. With this role, you can edit system configurations, inventory details, policies, and notification/integration settings. The RHEL operator role allows many of the RHEL administrator role functions, but it is restricted from editing compliance policies, content source templates, policies, or tasks. In addition, the RHEL operator role cannot execute remediation plans.

RHEL administrator: The *RHEL administrator* role provides comprehensive administrative privileges across your RHEL systems and Red Hat Lightspeed. With this role, you can manage system configurations, inventory, compliance policies, notifications, patch management, remediations, malware detection, and advisor recommendations. The role can also view and modify all vulnerability settings.



IMPORTANT

To use the consolidated roles effectively, you might need to remove the granular RHEL roles from the **Default access** group to prevent permission conflicts. This action automatically changes the name of the predefined **Default access** group to **Custom default access** group, after which, it is no longer automatically updated by Red Hat.

See *Predefined User Access roles* for a list of the roles included in the **Default admin access** group and a reference table that lists most of the predefined groups and roles that are available in the Red Hat Hybrid Cloud Console and the permissions included in each role.

Granular roles

The granular roles are specific roles for individual services that allow for fine-tuned control over specific features of Red Hat Lightspeed, for example, *Inventory Hosts administrator* or *Compliance viewer*. If you want to have more control over specific features of Red Hat Lightspeed and your use case does not align with the permissions included in the consolidated roles, use the granular predefined roles.

TIP

Across the Red Hat Lightspeed product documentation, the **Prerequisites** section for each procedure lists which predefined roles provide the permissions needed to use the features in that procedure. For example, if a procedure requires permissions to create and view remediations, the **Prerequisites** section for that procedure lists the *Remediations user* or other valid role as a recommended predefined role to use for that procedure.

Additional resources

- Preparing the Default access group for consolidated role permissions
- Predefined User Access roles

1.4.4. Check your permissions

Verify your current permissions and the roles or groups assigned to you in the Red Hat Hybrid Cloud Console. Check your permissions to troubleshoot access issues or understand your level of access to Red Hat Lightspeed applications.



NOTE

Only users with the Organization Administrator role can view the permissions of other users in the User Access settings and manage user permissions to Red Hat Lightspeed services. For more information, see the *Configure user permissions* section.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ My User Access**.
2. Optional: If you require additional permissions, use the Red Hat Hybrid Cloud Console Virtual Assistant to ask **"Contact my Organization Administrator"**. The assistant sends an email to the Organization Administrator on your behalf.

Verification

All of the applications that you have permissions to access are listed on this page and are grouped by product, for example, RHEL, OpenShift Container Platform, and Ansible Automation Platform.

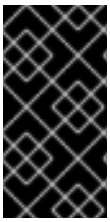
You can also filter your permissions by application, for example, by advisor, cost management, inventory, and remediations.

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC)
- Procedures for configuring User Access

1.4.5. Configure user permissions

If you are an Organization Administrator, you can view and manage user permissions for all users in your organization. Control access to Red Hat Lightspeed and other Red Hat Hybrid Cloud Console services through the User Access interface.



IMPORTANT

If you are not an Organization Administrator, you are unable to complete this task. However, you can check your own permissions for different applications by navigating to **Settings → Identity & Access Management→ My User Access**. Contact your Organization Administrator to request more permissions.

Prerequisites

- You have logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator, or you have the required administrator User Access role permissions.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ User Access**.

Verification

From here, you can create and manage:

- **Settings → Identity & Access Management→ User Access → Roles** to determine permissions to Red Hat Lightspeed services and features
- **Settings → Identity & Access Management→ User Access → Groups** to include one or more roles to align with a specific persona, job function, or team in your organization
- **Settings → Identity & Access Management→ User Access → Users** and their assignment to groups to inherit permissions from the roles assigned to those groups

Additional resources

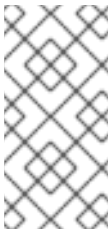
- User Access Configuration Guide for Role-based Access Control (RBAC)
- Procedures for configuring User Access
- Simplifying access management on Hybrid Cloud Console for Red Hat Lightspeed with predefined roles (Red Hat Knowledgebase)

1.4.6. User Access roles for permissions to compliance-service features

Understand the predefined roles that control access to the compliance service of Red Hat Lightspeed. Use these role definitions to assign appropriate permissions to users based on their responsibilities.

The following table shows the standard and enhanced access permissions provided by the predefined roles in User Access for the compliance service:

Table 1.1. Permissions provided by the User Access roles

User Access role	Grants permissions to ...	Included in the Default access group
Compliance administrator	• Manage compliance resources.	
Compliance viewer	• Read RHEL compliance reports.	X
RHEL administrator	• Do everything that a **RHEL operator** can do. • Administer RHEL system configs, inventory, compliance, notifications, patch management, execute remediation plans, malware detection, and advisor. • View and modify vulnerability settings.	
RHEL operator	• Do everything that a **RHEL viewer** can do. • Edit system configs, inventory, policies, notifications, and integrations. • View compliance reports, patch info, malware detections, and recommendations. • Create remediation plans, manage stale data, and change vulnerability settings.  **NOTE** The RHEL operator role is restricted from editing compliance policies, content source templates, policies, or tasks. Also, the RHEL operator role cannot execute remediation plans.	
RHEL viewer	• Read all available data across Red Hat Lightspeed services and features. ◦ View system configs, compliance reports, inventory data, patch info, vulnerabilities, and more to observe the state of resources or activities.  **NOTE** Cannot perform actions other than generating activation keys.	

1.5. ADDITIONAL RESOURCES

- Generating Compliance Service Reports

CHAPTER 2. GET STARTED WITH THE COMPLIANCE SERVICE

Learn how to configure your RHEL systems to report compliance data to the Red Hat Lightspeed application. You can install the Security Content Automation Protocol (SCAP) Security Guide (SSG) to perform the compliance scan.

2.1. CONFIGURE THE COMPLIANCE SERVICE

You can configure the compliance service and assign your RHEL systems to the policies. You can assign systems to policies by using either the Red Hat Lightspeed compliance service UI, or insights-client commands in the CLI.

Prerequisites

- The insights-client is deployed on the system.
- You have root privileges on the system.

Procedure

1. Check the version of RHEL that is on your system:

```
[user@rhlightspeed]$ cat /etc/redhat-release
```

2. Go to Red Hat Lightspeed Compliance - Supported configurations and find the supported SSG version for the RHEL minor version that is on your system.



NOTE

Some minor versions of RHEL support more than one version of SSG. The Red Hat Lightspeed compliance service will always show results for the latest supported version.

3. Verify that the supported version of the SSG package is installed on the system:
For example, for RHEL 8.4, run:

```
[root@rhlightspeed]# dnf info scap-security-guide-0.1.57-3.el8_4
```

4. If it is not installed, install the supported version of SSG on the system.
For example, for RHEL 8.4, run:

```
[root@rhlightspeed]# dnf install scap-security-guide-0.1.57-3.el8_4
```

5. Assign systems to policies by using the Red Hat Lightspeed compliance service UI, or by using insights-client commands in the CLI:
   - In the compliance service UI, navigate to Security > Compliance > SCAP policies and use one of the following methods to add systems:
     - Creating new SCAP policies
     - Editing included systems

- Or, in the CLI, add systems by using the following **insights-client** commands:
  - Run **insights-client --compliance-policies** to list available policies and their associated IDs
  - **insights-client --compliance-assign <ID>**
    For more information about using insights-client commands to add systems, see the following resources:
    - *Managing SCAP security policies in the Red Hat Lightspeed compliance service* in Assessing and monitoring security policy compliance of RHEL systems .
    - *Options for the insights-client* in Registering RHEL systems and configuring client tools with Red Hat Lightspeed.

6. After you add each system to its corresponding security policy, return to the system and run the compliance scan:

```
[root@rhlightspeed]# insights-client --compliance
```



NOTE

The scan can take one to five minutes to complete.

7. Navigate to Security > Compliance > Reports to view the results.
8. Optional: Use **cron** to schedule the compliance jobs to run.

Additional resources

- Red Hat Lightspeed Compliance - Supported configurations

2.2. SET UP RECURRING SCANS FOR RED HAT LIGHTSPEED SERVICES

You can manually scan and upload data collection reports on a regular schedule. Setting up these recurring scans gives you the most accurate recommendations from Red Hat Lightspeed services, such as compliance and malware detection.



IMPORTANT

Before you create a **cron** job, run all of the necessary commands manually to ensure that they are successful.

Prerequisites

- The Compliance and Malware Detection services are configured and running on your system.

Procedure

1. At the system prompt, issue the **crontab -e** command to edit the **crontab** file. This command opens your default text editor.

```
$ crontab -e
```

2. Add a **crontab** entry for the service that you want to run. For example:

```
10 20 * * * /bin/insights-client --compliance
10 21 * * * /bin/insights-client --collector malware-detection
```

3. Save the file and exit the text editor.

Results

In the preceding example, the first command uploads a compliance report to Red Hat Lightspeed every day at 20:10 local time. The second command uploads a malware detection report to Red Hat Lightspeed every day at 21:10 local time.

CHAPTER 3. MANAGE SCAP SECURITY POLICIES WITH THE RED HAT LIGHTSPEED COMPLIANCE SERVICE

You can create and manage your SCAP security policies in the compliance service UI. Define new policies, select the rules and systems that you want to associate with them, and edit existing policies when your requirements change.



IMPORTANT

Unlike most other Red Hat Lightspeed services, the compliance service does not run automatically on a default schedule. To upload OpenSCAP data to the Red Hat Lightspeed service, run **insights-client --compliance** either on-demand or on a scheduled job that you set.

3.1. CREATE NEW SCAP POLICIES

You must add each Red Hat Lightspeed registered system to one or more security policies before you can run a scan or see the results of a scan in the compliance service UI. You can also use filtering and search capabilities to refine the list of rules that you want to use in your policy.



IMPORTANT

If your RHEL servers span across multiple major releases of RHEL, you must create a separate policy for each major release. For example, all of your RHEL 7 servers must be on one *Standard System Security Profile for RHEL* policy. All of your RHEL 8 servers must be on another policy.

Procedure

1. Navigate to **Security > Compliance > SCAP Policies**
2. Click **Create new policy**.
3. On the **Create SCAP policy** page of the wizard, select the **RHEL major version** of the systems that you want to include in the policy.
4. Select one of the **policy types**, then click **Next**.
5. On the **Details** page, accept the name and description or enter your own.
6. Optional: Add a **Business objective** to give context, for example, “CISO mandate.”
7. Define a **compliance threshold** based on your requirements, then click **Next**.
8. Select the **Systems** that you want to include in this policy and click **Next**. The RHEL major version that you selected in the first step automatically determines which systems can be added to the policy.
9. Select the **Rules** that you want to include in each policy. Because each minor version of RHEL supports the use of a specific SCAP Security Guide (SSG) version, the rule set for each RHEL minor version is slightly different. If there is more than one, we use the latest.

10. Optional: Use the filtering and search capabilities to refine the list of rules. For example, to show only the highest severity rules, click the primary filter drop-down and select **Severity**. In the secondary filter, select **High** and **Medium**.

11. Repeat the preceding steps for *each* RHEL minor version tab.
The rules displayed by default are designated for the policy type and version of SSG. By default, the **Selected only** toggle is selected.

Next steps

1. After you select rules for each Red Hat Enterprise Linux minor version SSG, click **Next**.
2. On the **Review** page, verify that the information is correct, then click **Finish**.
3. Wait for the app to create the policy, then click **Return to application** to view your new policy.



NOTE

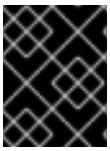
On the system, run the compliance scan so that your results display in the compliance service UI.

3.2. EDIT COMPLIANCE POLICIES IN THE RED HAT LIGHTSPEED UI

You can edit policies in the Red Hat Lightspeed UI. You can change the policy details or which rules and systems are included.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with at least one of the following roles:
  - **Compliance administrator**
  - **RHEL administrator**



IMPORTANT

The required roles include enhanced permissions that are not granted by default to all Red Hat Lightspeed users.

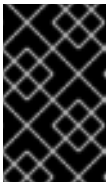
Procedure

1. Navigate to the **Security > Compliance > SCAP policies** page.
2. Locate the policy that you want to edit.
3. To edit **Compliance threshold**, **Business objective**, and **Policy description**:
   a. Click the policy name that you want to modify.
   b. In the **Policy details** view, edit the relevant fields and then click the checkmark to the right of the fields to save the changes.
4. To edit the policy rules:

a. On the right side of the policy row, click the **More actions** icon, , then click **Edit policy**.

b. In the **Edit** pop-up, click the **Rules** tab.

c. Select a RHEL minor version.



IMPORTANT

Because a different SCAP Security Guide (SSG) version exists for each minor version of RHEL, you must edit the rules for each minor version of RHEL separately.

d. Search for the rule name or its identifier to locate the rules that you want to remove.

e. Select the rules that you want to add. Clear any rules that you want to remove.

f. Repeat the preceding steps for each RHEL minor version tab.

g. Click **Save**.

5. To edit included systems:

a. On the right side of the policy row, click the **More actions** icon, , and click **Edit policy**.

b. In the **Edit** pop-up, click **Systems**. A list of all available systems is displayed.

c. Select systems that you want to add. Clear any systems that you want to remove.

Verification

1. Navigate to the **Security > Compliance > SCAP policies** page and select the policy that you edited.
2. Verify that the edits were successful.

3.3. COMPLIANCE API CALLS FOR EDITING POLICY DETAILS

The Compliance API supports the GET, POST, PATCH, and DELETE HTTP methods to edit your policy details. The API works asynchronously.

The API offers the following services for Compliance policies

- **GET /policies**
- **POST /policies**
- **DELETE /policies/{policy_id}**
- **GET /policies/{policy_id}**
- **PATCH /policies/{policy_id}**

- **GET /policies/{policy_id}/systems**
- **DELETE /policies/{policy_id}/systems/{system_id}**
- **PATCH /policies/{policy_id}/systems{system_id}**
- **GET /policies/{policy_id}/tailorings**
- **GET /policies/{policy_id}/tailorings/{tailoring_id}**
- **PATCH /policies/{policy_id}/tailorings/{tailoring_id}**
- **GET /policies/{policy_id}/tailorings/{tailoring_id}/rules**
- **DELETE /policies/{policy_id}/tailorings/{tailoring_id}/rules/{rule_id}**
- **PATCH /policies/{policy_id}/tailorings/{tailoring_id}/rules/{rule_id}**
- **GET /policies/{policy_id}/tailorings/{tailoring_id}/tailoring_file.json**

Successful requests return the following responses

- **200 — Success**
- **202 — Successfully deleted (for the DELETE method)**
- **202 — Successfully updated (for the PATCH method)**

The API works asynchronously. To get a list of policy details, submit the **GET /policies/{policy_id}** request to the Compliance API. Use the **PATCH /policies/{policy_id}** request to update the details.

Additional resources

- Using APIs to configure Red Hat Lightspeed services
- API Catalog and Documentation: Compliance API v2
- Red Hat Lightspeed API Cheat Sheet

3.4. EDIT THE POLICY DETAILS BY USING THE COMPLIANCE API

You can use the REST API entry point to change policy details or to include and exclude rules and systems. There are multiple ways to make API calls to the Red Hat Lightspeed APIs.

There are multiple ways to make API calls to the Red Hat Lightspeed APIs. For more information, refer to Using APIs to configure Red Hat Lightspeed services .

For more information about the various ways to make calls to the Red Hat Lightspeed APIs, see Make API calls.

The example procedure in this section uses the API Catalog at developers.redhat.com to edit policy details. Before you edit Compliance policy details, you can use the **GET /policies** API call to query the API for the complete list of policy IDs and details.

Prerequisites

Prerequisites

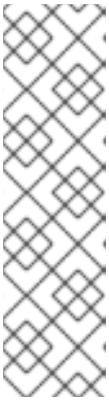
- You have the programming language that you want to use installed on your system.
- You have an access token that you obtained from your service account or from an offline access token.
- You have a service account that is configured with necessary permissions. To create and configure a service account, see Creating and managing a service account .
- You have the **policy_id** for the policy that you want to edit.

Procedure

1. Open the API catalog in a web browser. The API catalog lists the APIs that are available for Red Hat Lightspeed and other services.
2. From the catalog, select **Compliance** and then click **v2**. The Cloud Services for RHEL Compliance API v2 page opens. The page lists the base URL of the Compliance API and the requirements for authentication.
3. Under the **Policies** list, click the drop-down next to **Update a Policy**. The description for the operation includes parameters that you can use to refine your API call and expected responses from the server. In addition, the panel on the right side of the page generates an example API call for the PATCH operation in multiple programming languages.
4. In the panel, click the drop-down and select a language. The panel displays sample code for the PATCH operation in the language you selected.
5. Copy the sample code and paste it into a code file where you want to call the PATCH command. For example, in Python:

```
import requests
url = "https://console.redhat.com/policies/%7Bpolicy_id%7D"
headers = {"Content-Type": "application/vnd.api+json"}
response = requests.patch(url, headers=headers)
print(response.json())
```

6. Replace **%7Bpolicy_id%7D** with your **policy_id**
7. Paste the update that you want to make in **application/vnd.api+json**. The update takes the form **object: value**. For example, paste **compliance_threshold: 50** to change the compliance threshold to 50.



NOTE

To see the fields that you can update for the policy, click the **policy_update** link under **Request Body Schema** next to the sample code. The link shows the following available fields and their expected values:

- **business_objective**
- **compliance_threshold**
- **description**.

8. Select **Enter** to run your code.

Verification

If the server returns a response of 202, your API call was successful. If the code returns a response other than 202, refer to the error message included with the response code to determine how the API call failed and how to remedy the reason for the failure.

Additional resources

- Red Hat Lightspeed API Catalog and Documentation
- https://console.redhat.com/api/compliance/v2
- Red Hat Developers API Catalog
- Creating and managing a service account

3.5. VIEW SCAP POLICIES

After you register your system to Red Hat Lightspeed, you can view all available compliance policies for that system by using the **insights-client --compliance-policies** command. This command enables you to view the **ID** and **Title** of each policy and whether they are **Assigned**.

Prerequisites

- The insights-client is installed on the system.
- You are logged in to a system where you have root privileges.

Procedure

- In the command line, enter:

```
[root@rhlightspeed]# insights-client --compliance-policies
```

Result

A list of compliance policies that are supported for the system is displayed. The output includes the **ID** and **Title** for the policies and whether the policies are **Assigned**, indicated by either **TRUE** or **FALSE**.

Additional resources

- Registering RHEL systems and configuring client tools with Red Hat Lightspeed

3.6. ASSIGN SYSTEMS TO SCAP POLICIES

You can assign (add) systems to SCAP policies by using the **insights-client --compliance-assign** command. This command enables you to create custom automation for your systems and for the SCAP policies that are available to those systems.

Prerequisites

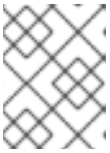
- The insights-client is installed on the system.

- You are logged in to a system where you have root privileges.
- You ran the **insights-client --compliance-policies** command.

Procedure

1. In the command line, enter:

```
[root@rhlightspeed]# insights-client --compliance-assign <ID>
```



NOTE

Use a policy ID from the **insights-client --compliance-policies** command output.

Verification

1. Navigate to Security > Compliance > SCAP policies .
2. Click the name of the policy that you assigned the system to.
3. Click the **Systems** tab. The system is listed for the policy.
4. Optional: Run the **insights-client --compliance-policies** command to confirm that the **Assigned** value is set to **True** for the policy.

Additional resources

- Registering RHEL systems and configuring client tools with Red Hat Lightspeed

3.7. UNASSIGN SYSTEMS FROM SCAP POLICIES

You can use the **insights-client** to unassign (remove) systems or to automate steps in your SCAP policies.

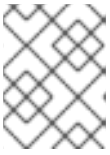
Prerequisites

- The insights-client is installed on the system.
- You are logged in to a system where you have root privileges.
- You ran the **insights-client --compliance-policies** command.

Procedure

- On the command line, enter

```
[root@rhlightspeed]# insights-client --compliance-unassign <ID>
```



NOTE

Use a policy ID from the **insights-client --compliance-policies** command output.

Verification

1. Navigate to Security > Compliance > SCAP policies .
2. Click the name of the policy that you assigned the system to.
3. Click the **Systems** tab. The system is no longer listed.
4. Optional: Run the **insights-client --compliance-policies** command to confirm that the **Assigned** value is set to **False** for the policy.

Additional resources

- Registering RHEL systems and configuring client tools with Red Hat Lightspeed

3.8. VIEW POLICY RULES

Red Hat Lightspeed Compliance displays rules in categorized groups so that similar rules display together. You can see rules grouped by category or classification for the compliance checks that will take place for a policy.

The nested group structure, called the tree view, is the default view. The tree view provides additional contextual information, like categories of rules and multiple rules for a policy, when applicable. You can also see rules that have editable values.

You can view rules in the tree view or the classic view. In the classic view, rules appear in a linear list. Switch from the tree view ( ) to the classic view ( ) by toggling between the two buttons under **View policy rules**.



NOTE

- When you use the filter feature to search for a specific rule, the view automatically switches to the classic view.
- After you expand a rule to show additional information, it will stay in the expanded view, even if you switch to a different view.

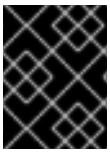
Additional resources

- Editing compliance policies
- Creating new SCAP policies
- Generating Compliance Service Reports

CHAPTER 4. ANALYZING AND TRIAGING YOUR COMPLIANCE REPORTS

The compliance service displays data for each policy and system that is registered to the service, producing a large amount of data. Learn how to refine the bulk of compliance service data in Reports, SCAP policies, and Systems to focus on the most important systems and policies.

The compliance service enables users to set filters on lists of systems, rules, and policies. Like other Red Hat Lightspeed services, the compliance service also enables filtering by system-group tags. However, because compliance-registered systems use a different reporting mechanism, the tag filters must be set directly in lists of systems in the compliance UI views, not in the global **Filter by status** drop-down.



IMPORTANT

To see accurate data for your systems, always run **insights-client --compliance** on each system before you view the results in the UI.

4.1. COMPLIANCE REPORTS

To focus on a specific set of reports, you can use primary and secondary filters. These filters enable you to search with narrow criteria, making it easier to find relevant reports.

From Security > Compliance > Reports , you can use the following filters:

- **Policy name**: Search for a policy by name
- **Policy type**: Select from the policy types that are configured for your infrastructure in the compliance service
- **Operating system**: Select one or more RHEL operating system major versions
- **Systems meeting compliance**: Show policies that have a certain percentage (range) of compliant systems

4.2. SCAP POLICIES

You can filter by names in Security > Compliance > SCAP policies to locate a specific policy.

From Security > Compliance > SCAP policies , click the policy name to see the policy card which includes the following information:

- **Details**: View details such as compliance threshold, business objective, operating system, and SSG version.
- **Rules**: View and filter the rules included in the specific SSG version of the policy by **Name**, **Severity** and **Remediation available**. Then sort the results by **Rule name**, **Severity** or **Ansible Playbook support**.
- **Systems**: Search by system name to locate a specific system that is associated with the policy. Click the system name to see more information and any issues.

4.3. OVERVIEW OF SYSTEMS IN THE COMPLIANCE SERVICE

The default functionality on Security > Compliance > Systems is to search by system name.

You can search for system names by using the following filters:

- **Tags**: Search by system group or tag name
- **Name**: Search by system name
- **Policy**: Search by policy name and see the systems included in that policy
- **Operating system**: Search by RHEL operating system major versions to see only RHEL 7 or RHEL 8 systems

4.4. SEARCH OPTIONS IN THE COMPLIANCE SERVICE

The search function in the compliance service works in the context of the page that you are viewing. You can search for policies, systems, and rules.

You can use the following search functions:

- **SCAP Policies**: Search for a specific policy by name
- **Systems**: Search by system name, policy, or Red Hat Enterprise Linux operating system major version
- **Rules list (single system)**: Search by the rule name or identifier

CHAPTER 5. ORGANIZE SYSTEMS IN RED HAT LIGHTSPEED BY USING FILTERING AND TAGS

Filter and organize your systems in Red Hat Lightspeed by using SAP workload tags, Satellite host group tags, and custom tags. Tagging enables you to focus on specific system groups across multiple services, which simplifies management for large-scale deployments.

5.1. FILTERS AND TAGS FOR SAP WORKLOADS

Red Hat Lightspeed automatically tags systems by SAP workload type and SAP ID (SID), which you can use to filter and organize SAP systems.

Use these automatic tags to filter and organize your SAP systems throughout the Red Hat Lightspeed application. No customization or manual tagging is required. To filter to view your SAP workloads, use **Workloads > SAP**.

5.2. FILTERS AND TAGS FOR SATELLITE HOST GROUPS

Satellite host groups are configured in Satellite and are automatically recognized by Red Hat Lightspeed.

5.3. CUSTOM SYSTEM TAGGING

Apply custom tags to your registered systems to add contextual markers and organize your Red Hat Lightspeed inventory. With custom tags, you can filter and focus on related groups of systems, which simplifies management for large-scale deployments.

You can also apply predefined tags to your systems. The advisor service uses predefined tags to generate targeted recommendations for systems that require specific attention, such as those with high security needs.

5.3.1. Filter structure

Filters use a **namespace=value** or **key=value** paired structure.

- **Namespace.** The namespace is the name of the ingestion point, *insights-client*. This value cannot be changed. The **tags.yaml** file is abstracted from the namespace, which is injected by the client before upload.
- **Key.** You can create the key or use a predefined key from the system. You can use a mix of capitalization, letters, numbers, symbols, and whitespace.
- **Value.** You can define your own descriptive string value. You can use a mix of capitalization, letters, numbers, symbols, and whitespace.

5.3.2. Create a custom group and the tags.yaml file

To create and add tags to **/etc/insights-client/tags.yaml**, use **insights-client** with the **--group=<name-you-choose>** option. Running this command on your systems that are registered to Red Hat Lightspeed completes the following actions:

- Creates the **etc/insights-client/tags.yaml** file

- Adds the **group=** key and **<name-you-choose>** value to **tags.yaml**
- Uploads a fresh archive from the system to the Red Hat Lightspeed service, making the new tag immediately visible along with your latest results

Prerequisites

- Root-level access to your system.

Procedure

1. Run the following command as root, adding your custom group name in place of **<name-you-choose>**:

```
[root@server ~]# insights-client --group=<name-you-choose>
```

2. Optional: To add additional tags, edit the **/etc/insights-client/tags.yaml** file.
3. Navigate to **Inventory → Systems** and log in if necessary.
4. Click the **Filter by tags** drop-down menu. You can also use the search box to enter all or part of the tag's name to automatically show systems with that text in the tags.
5. Scroll up or down the list to locate the tag.
6. Click the tag to filter by it.
7. Verify that your system is among the results on the advisor systems list.
   a. Navigate to **Inventory → Systems** and log in if necessary.
   b. Activate the **Name** filter and begin typing the system name until you see your system, then select it.
   c. The tag symbol is a darker color, and the number beside it shows the correct number of tags applied.

5.3.3. Add and modify custom system tags by editing tags.yaml

Edit the **tags.yaml** file to add or modify custom system tags after creating a **group** tag. Using custom system tags can help you streamline the organization and filtering of systems when you use Red Hat Lightspeed inventory and other services.

Prerequisites

- Root-level access to your system.

Procedure

1. Open the tag configuration file, **tags.yaml**, in an editor.

```
[root@server ~]# vim /etc/insights-client/tags.yaml
```

2. Edit the file contents or add **key=value** pairs as needed. Use a mix of capitalization, letters, numbers, symbols, and whitespace. Example **tags.yaml** with multiple tags:

```
group: _group-name-value_
location: _location-name-value_
description:
- RHEL8
- SAP
key 4: value
```

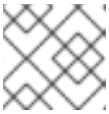
3. Save your changes and close the editor.

4. Initiate an upload to Red Hat Lightspeed.

```
[root@server ~]# insights-client
```

5. Navigate to **Inventory → Systems** and log in if necessary.

6. In the **Filter Results** box, click the down arrow and select one of the filters or enter the name of the filter and select it.



NOTE

You can search by the tag key or by its value.

7. Find your system among the results.

8. Verify that the filter icon is highlighted and shows a number representing the number of filters applied to the system.

5.4. PREDEFINED TAGS

In addition to custom tags, Red Hat Lightspeed provides predefined tags that the advisor service can use to create targeted recommendations. These predefined tags are useful for systems that require more security or need different networking performance levels.

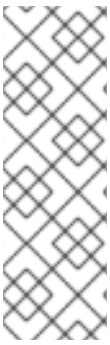
To get the extended security hardening and enhanced detection and remediation capabilities offered by predefined tags, opt in and configure the tags. After configuration, the advisor service provides recommendations based on tailored severity levels and the preferred network performance that applies to your systems.

Use the **/etc/insights-client/tags.yaml** file to tag systems with predefined tags, similar to how you tag systems in the inventory service. The predefined tags are configured using the same **key=value** structure used to create custom tags. Details about the Red Hat-predefined tags are in the following table:

Table 5.1. List of supported predefined tags

Key	Value	Note

Key	Value	Note
security	**normal** (default) / **strict**	With the **normal** (default) value, the advisor service compares the system's risk profile to a baseline that comes from the default configuration of the most recent version of RHEL and from often-used usage patterns. This option keeps recommendations focused, actionable, and low in numbers. The **strict** value is best for systems that are security-sensitive. This option uses a stricter baseline and can show recommendations even on the most up-to-date RHEL installations.
network_performance	**null** (default) / **latency** / **throughput**	The network performance values, either latency or throughput according to your business requirement, affect the severity of an advisor service recommendation to a system.



NOTE

The names of predefined tag keys are reserved for specific use. If you already use the key **security**, with a value that differs from one of the predefined values, you do not see a change in your recommendations. You only see a change in recommendations if your existing **key=value** is the same as one of the predefined keys. For example, if you have a **key=value** of **security: high**, your recommendations do not change because of the predefined tags that Red Hat has already defined. If you currently have a **key=value** pair of **security: strict**, you see a change in the recommendations for your systems.

Additional resources

- Using system tags to enable extended security hardening recommendations
- Leverage tags to make Red Hat Lightspeed Advisor recommendations understand your environment better
- Custom system tagging

5.4.1. Configure predefined tags

Configure predefined tags to adjust recommendations and gain extended security hardening, enhanced detection, and remediation capabilities within the Red Hat Lightspeed advisor service.

Prerequisites

- You have root-level access to your system.
- You have insights-client installed.
- You have systems registered within the insights-client.
- You have created the **tags.yaml** file. For information about creating the **tags.yaml** file, see Configure custom group tags.

Procedure

1. Use the command line and your preferred editor to open **/etc/insights-client/tags.yaml**. (The following example uses Vim.)

```
[root@server ~]# vi /etc/insights-client/tags.yaml
```

2. Edit the **/etc/insights-client/tags.yaml** file to add the predefined **key=value** pair for the tags. This example shows how to add **security: strict** and **network_performance: latency** tags.

```
# cat /etc/insights-client/tags.yaml
group: redhat
location: Brisbane/Australia
description:
- RHEL8
- SAP
security: strict
network_performance: latency
```

3. Save your changes.
4. Close the editor.
5. Optional: Run the **insights-client** command to generate an upload to Red Hat Lightspeed, or wait until the next scheduled Red Hat Lightspeed upload.

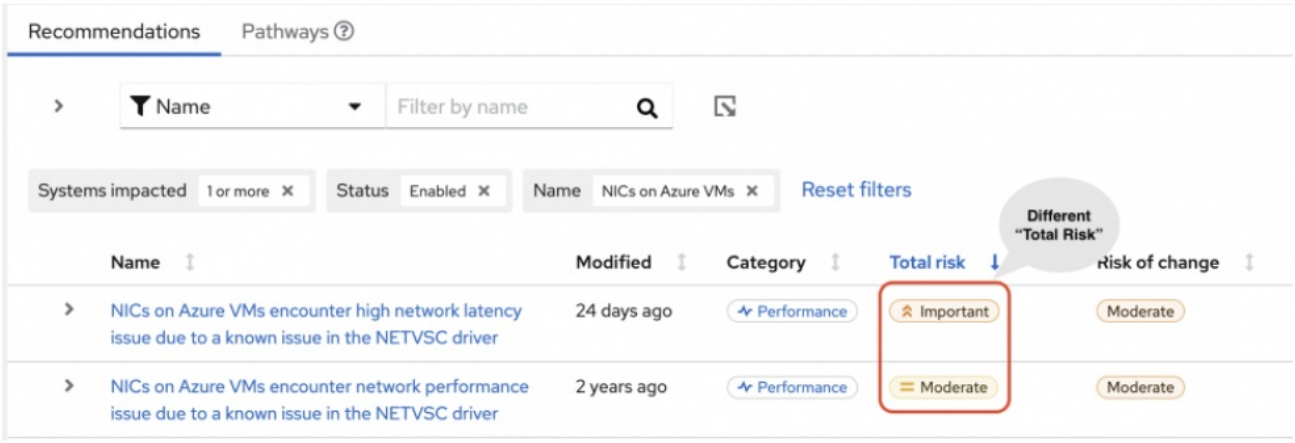
```
[root@server ~]# insights-client
```

Verification

After generating an upload to Red Hat Lightspeed (or waiting for the next scheduled Red Hat Lightspeed upload), you can discover whether the tags are in the production environment by accessing Red Hat Lightspeed > RHEL > Inventory > Systems . Find your system and look for the newly created tags. A table displays:

- Name
- Value
- Tag Source (for example, insights-client).

The following image of the advisor service shows a system with the **network_performance: latency** tag configured.



The system shows a recommendation with a higher Total Risk level of Important. The system without the **network_performance: latency** tag has a Total Risk of Moderate. You can make decisions about prioritizing the system with higher Total Risk.

5.5. ADDITIONAL RESOURCES

- Refining your view of systems in inventory
- Enable SQL Server recommendations in Advisor
- Group and tag filters in the compliance service

5.6. GROUP AND TAG FILTERS IN THE COMPLIANCE SERVICE

The compliance service enables users to apply tag and group filters to systems that report compliance data. You can set these filters by using the primary and secondary filters in the compliance service UI.

You must set tag and group filters by using the primary and secondary filters located above lists of systems in the compliance service UI. This is because, unlike most of the other services in the Red Hat Lightspeed application, the compliance service only shows data for systems under the following conditions:

- The system is associated with a compliance service security policy.
- The system is reporting compliance data to Red Hat Lightspeed using the **insights-client --compliance** command.

CHAPTER 6. REFERENCE MATERIALS

To learn more about the compliance service, see the following resources:

- Generating Compliance Service Reports
- Red Hat Lightspeed Documentation
- Red Hat Lightspeed Product Support page